

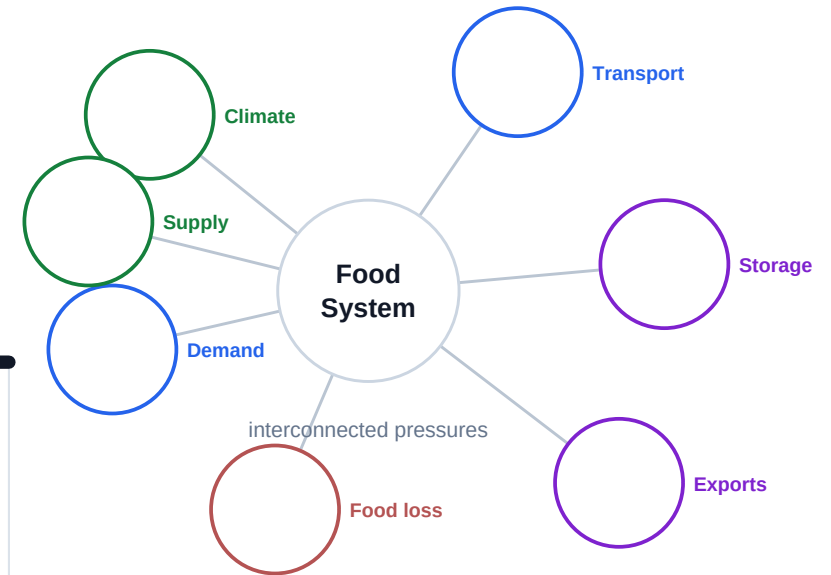
Systems Under Pressure

The Mathematics of Food Systems

"All models are wrong, but some are useful."

George Box

A spacious mathematical field notebook for sketching, modeling, revising, and explaining.



Rough work belongs here.

This journal is not a worksheet packet. It is the visible record of your group's evolving mathematical thinking.

Value uncertainty

Label assumptions, questions, disagreements, and places where a model stops making sense.

Revise visibly

Cross out, annotate, draw arrows, compare versions, and explain why your thinking changed.

Field note style

Messy is acceptable. Hidden thinking is not. The strongest journals show graph, algebra, context, units, limitations, and revision together.

Use the journal to make standards visible.

The rubric does not only score final answers. It looks for reasoning, interpretation, revision, limitations, precision, and flexible thinking across the investigation.

1. Sequences and Series

Content Standard

2. Exponentials and Logarithms

Content Standard

3. Polynomials

Content Standard

4. Mathematical Communication and Interpretation

Skill Grade

5. Comparing Models, Assumptions, and Limitations

Skill Grade

6. Algebraic Precision and Flexible Number Sense

Skill Grade

Entering the System

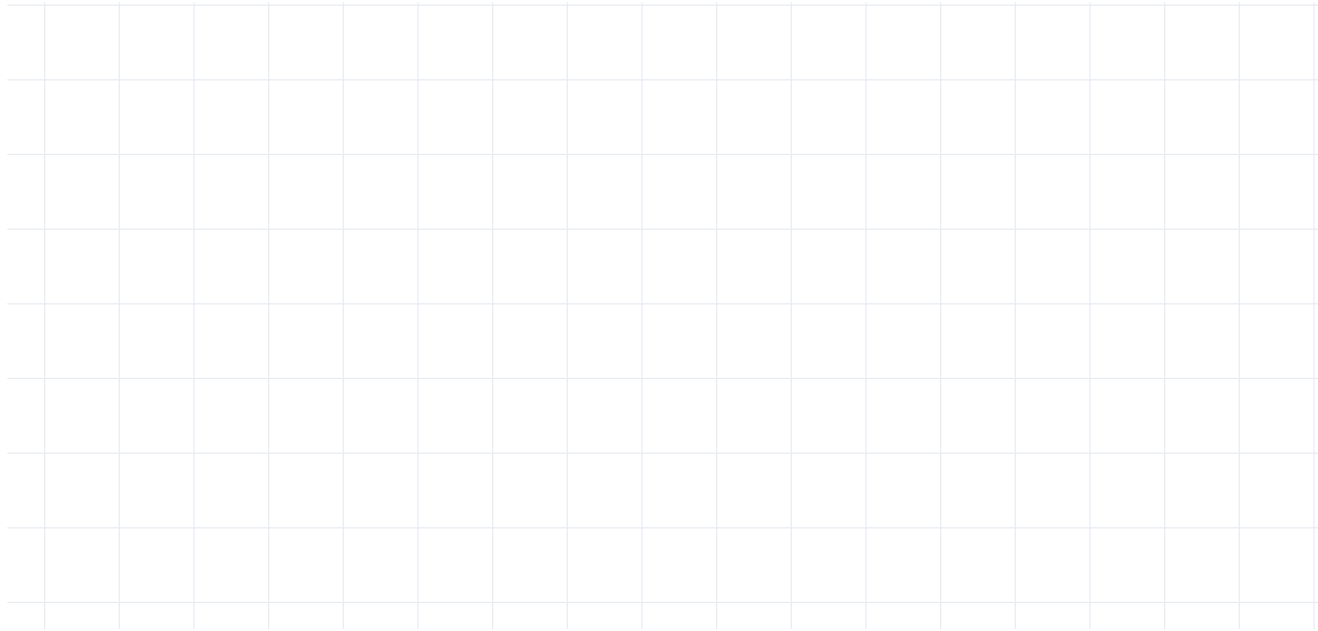
What relationships are we beginning to notice?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Sketch variables, links, pressures, unknowns, and possible feedback loops.

student thinking space



Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Variable | Units | Why it matters | What is hard to measure

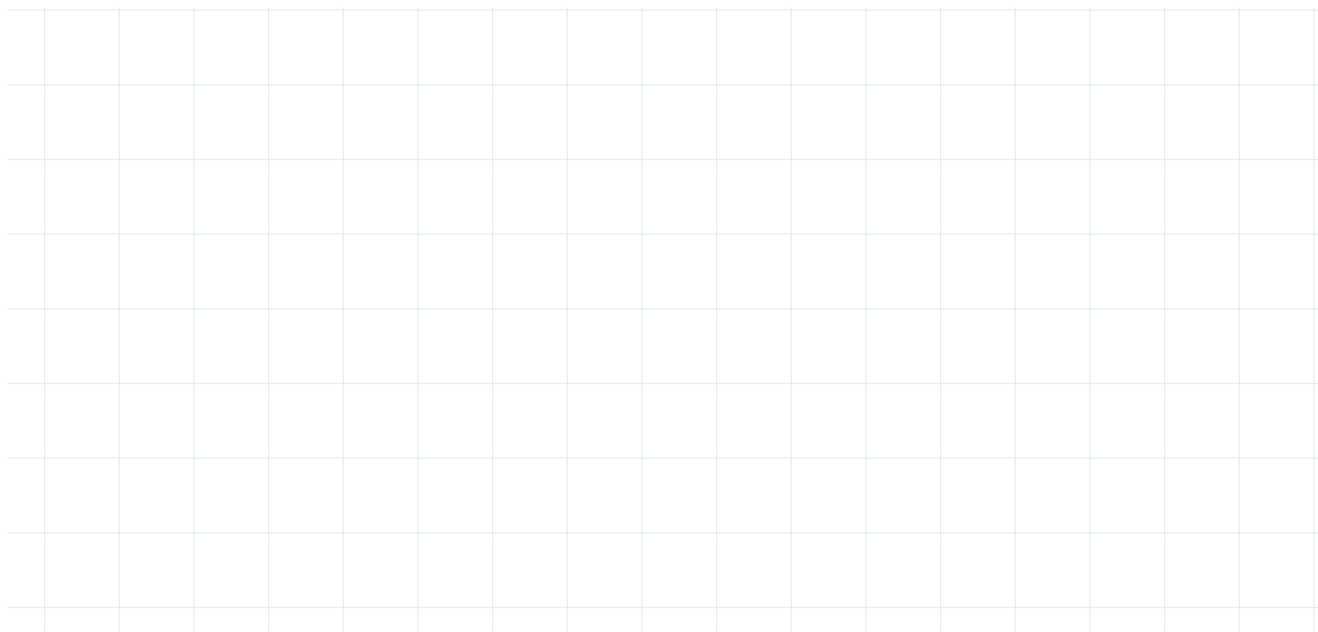
student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Predict the shape of one relationship before using Desmos, Sheets, or another tool.

student thinking space

A large grid for student thinking space, consisting of 20 columns and 15 rows of squares.

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Patterns noticed, uncertainties, assumptions, important variables.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Modeling Change

How is the system changing over time?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Why is a sequence appropriate? What would change in a continuous model?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Use evidence. Explain what each term represents in context.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Write both formulas. Annotate every variable and parameter.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

What does the cumulative sum represent in the real system?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

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Recalculation, comparison, graph interpretation, or revised assumption.

student thinking space

Reflection prompt

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Acceleration, Decay, and Thresholds

When does repeated change become accelerated or compounding?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Why does exponential reasoning fit better than repeated addition?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

How does continuous exponential modeling change the interpretation compared to discrete terms?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

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Show algebraic steps, graph verification, units, and interpretation.

student thinking space

Reflection prompt

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Compare 3%, 5%, and 8% growth or decay. What happens to thresholds?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Where does exponential behavior stop being realistic? What hidden variables matter?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Constraints, Capacity, and Cost

How do physical and economic constraints reshape the system?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Height, interior dimensions, usable space, loading floor.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Factored form, standard form, and why multiplying dimensions creates a cubic.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

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Degree, even/odd, leading coefficient, as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

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Change dimensions, cost, refrigeration, transport, or available space. Compare results.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Revision, Synthesis, and Communication

What story does our mathematics tell?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Lens | What did it reveal? | Where did it fail?

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Compare mathematical lenses and assumptions.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Use evidence, assumptions, limitations, tradeoffs, and implications.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Revision, hidden-variable analysis, limitation critique, or r/R^2 interpretation.

student thinking space

Reflection prompt

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?